5. A map $p:E\to B$ with B not necessarily path-connected is defined to be a quasifibration if the following equivalent conditions are satisfied:

(i) For all $b\in B$ and $x_0\in p^{-1}(b)$, the map $p_*:\pi_i(E,p^{-1}(b),x_0)\to\pi_i(B,b)$ is an isomorphism for $i>0$ and $\pi_0(p^{-1}(b),x_0)\to\pi_0(E,x_0)\to\pi_0(B,b)$ is exact.

(ii) The inclusion of the fiber $p^{-1}(b)$ into the homotopy fiber F_b of p over b is a weak homotopy equivalence for all $b\in B$.

(iii) The restriction of p over each path-component of B is a quasifibration according to the definition in this section.

Show these three conditions are equivalent, and prove Lemma 4K.3 for quasifibrations over non-pathconnected base spaces.

6. Let X be a complex of spaces over a Δ-complex Γ, as defined in §4.G. Show that the natural projection $\Delta X\to\Gamma$ is a quasifibration if all the maps in X associated to edges of Γ are weak homotopy equivalences.

4.L Steenrod Squares and Powers

The main objects of study in this section are certain homomorphisms called **Steenrod squares** and **Steenrod powers**:

$$Sq^i:H^n(X;\mathbb{Z}_2)\to H^{n+i}(X;\mathbb{Z}_2)$$
$$P^i:H^n(X;\mathbb{Z}_p)\to H^{n+2i(p-1)}(X;\mathbb{Z}_p)\quad\text{for odd primes } p$$

The terms 'squares' and 'powers' arise from the fact that Sq^i and P^i are related to the maps $\alpha\mapsto\alpha^2$ and $\alpha\mapsto\alpha^p$ sending a cohomology class α to the 2-fold or p-fold cup product with itself. Unlike cup products, however, the operations Sq^i and P^i are stable, that is, invariant under suspension.

The operations Sq^i generate an algebra $\mathcal{A}_2$, called the Steenrod algebra, such that $H^*(X;\mathbb{Z}_2)$ is a module over $\mathcal{A}_2$ for every space X, and maps between spaces induce homomorphisms of $\mathcal{A}_2$-modules. Similarly, for odd primes p, $H^*(X;\mathbb{Z}_p)$ is a module over a corresponding Steenrod algebra $\mathcal{A}_p$ generated by the P^i's and Bockstein homomorphisms. Like the ring structure given by cup product, these module structures impose strong constraints on spaces and maps. For example, we will use them to show that there do not exist spaces X with $H^*(X;\mathbb{Z})$ a polynomial ring $\mathbb{Z}[\alpha]$ unless α has dimension 2 or 4, where there are the familiar examples of $\mathbb{C}\mathrm{P}^\infty$ and $\mathbb{H}\mathrm{P}^\infty$.

This rather lengthy section is divided into two main parts. The first part describes the basic properties of Steenrod squares and powers and gives a number of examples and applications. The second part is devoted to constructing the squares and powers and showing they satisfy the basic properties listed in the first part. More extensive applications will be given in [SSAT] after spectral sequences have been introduced. Most applications of Steenrod squares and powers do not depend on how these operations are actually constructed, but only on their basic properties. This is similar to the situation for ordinary homology and cohomology, where the axioms generally suffice for most applications. The construction of Steenrod squares and powers and the verification of their basic properties, or axioms, is rather interesting in its own way, but does involve a certain amount of work, particularly for the Steenrod powers, and this is why we delay the work until later in the section.

We begin with a few generalities. A **cohomology operation** is a transformation $\Theta = \Theta_X : H^m(X;G) \to H^n(X;H)$ defined for all spaces X, with a fixed choice of m, n, G, and H, and satisfying the naturality property that for all maps $f : X \to Y$ there is a commuting diagram as shown at the right. For example, with coefficients in a ring R the transformation $H^m(X;R) \to H^{mp}(X;R)$, $\alpha \mapsto \alpha^p$, is a cohomology operation since $f^*(\alpha^p) = (f^*(\alpha))^p$. Taking $R = \mathbb{Z}$, this example shows that cohomology operations need not be homomorphisms. On the other hand, when $R = \mathbb{Z}_p$ with p prime, the operation $\alpha \mapsto \alpha^p$ is a homomorphism. Other examples of cohomology operations we have already encountered are the Bockstein homomorphisms defined in §3.E. As a more trivial example, a homomorphism $G \to H$ induces change-of-coefficient homomorphisms $H^m(X;G) \to H^m(X;H)$ which can be viewed as cohomology operations.

$$\begin{array}{ccc} H^m(Y;G) & \xrightarrow{\Theta_Y} & H^n(Y;H) \\ \big\downarrow f^* & & \big\downarrow f^* \\ H^m(X;G) & \xrightarrow{\Theta_X} & H^n(X;H) \end{array}$$

In spite of their rather general definition, cohomology operations can be described in somewhat more concrete terms:

Proposition 4L.1. *For fixed m, n, G, and H there is a bijection between the set of all cohomology operations $\Theta : H^m(X;G) \to H^n(X;H)$ and $H^n(K(G,m);H)$, defined by $\Theta \mapsto \Theta(\iota)$ where $\iota \in H^m(K(G,m);G)$ is a fundamental class.*

Proof: Via CW approximations to spaces, it suffices to restrict attention to CW complexes, so we can identify $H^m(X;G)$ with $\langle X, K(G,m)\rangle$ as in Theorem 4.57. If an element $\alpha \in H^m(X;G)$ corresponds to a map $\varphi : X \to K(G,m)$, so $\varphi^*(\iota) = \alpha$, then $\Theta(\alpha) = \Theta(\varphi^*(\iota)) = \varphi^*(\Theta(\iota))$ and Θ is uniquely determined by $\Theta(\iota)$. Thus $\Theta \mapsto \Theta(\iota)$ is injective. For surjectivity, given an element $\alpha \in H^n(K(G,m);H)$ corresponding to a map $\theta : K(G,m) \to K(H,n)$, then composing with θ defines a transformation $\langle X, K(G,m)\rangle \to \langle X, K(H,n)\rangle$, that is, $\Theta : H^m(X;G) \to H^n(X;H)$, with $\Theta(\iota) = \alpha$. The naturality property for Θ amounts to associativity of the compositions

$$X \xrightarrow{f} Y \xrightarrow{\varphi} K(G,m) \xrightarrow{\theta} K(H,n)$$

and so Θ is a cohomology operation. $\square$

A consequence of the proposition is that cohomology operations cannot decrease dimension, since $K(G,m)$ is $(m-1)$-connected. Moreover, since $H^m(K(G,m);H) \approx$

$\mathrm{Hom}(G,H)$, it follows that the only cohomology operations that preserve dimension are given by coefficient homomorphisms.

The Steenrod squares $Sq^i : H^n(X;\mathbb{Z}_2) \to H^{n+i}(X;\mathbb{Z}_2)$, $i \ge 0$, will satisfy the following list of properties, beginning with naturality:

(1) $Sq^i(f^*(\alpha)) = f^*(Sq^i(\alpha))$ for $f : X \to Y$.
(2) $Sq^i(\alpha + \beta) = Sq^i(\alpha) + Sq^i(\beta)$.
(3) $Sq^i(\alpha \smile \beta) = \sum_j Sq^j(\alpha) \smile Sq^{i-j}(\beta)$ (the Cartan formula).
(4) $Sq^i(\sigma(\alpha)) = \sigma(Sq^i(\alpha))$ where $\sigma : H^n(X;\mathbb{Z}_2) \to H^{n+1}(\Sigma X;\mathbb{Z}_2)$ is the suspension isomorphism given by reduced cross product with a generator of $H^1(S^1;\mathbb{Z}_2)$.
(5) $Sq^i(\alpha) = \alpha^2$ if $i = |\alpha|$, and $Sq^i(\alpha) = 0$ if $i > |\alpha|$.
(6) $Sq^0 = \mathbb{1}$, the identity.
(7) Sq^1 is the $\mathbb{Z}_2$ Bockstein homomorphism β associated with the coefficient sequence $0 \to \mathbb{Z}_2 \to \mathbb{Z}_4 \to \mathbb{Z}_2 \to 0$.

The first part of (5) says that the Steenrod squares extend the squaring operation $\alpha \mapsto \alpha^2$, which has the nice feature of being a homomorphism with $\mathbb{Z}_2$ coefficients. Property (4) says that the Sq^i's are stable operations, invariant under suspension. The actual squaring operation $\alpha \mapsto \alpha^2$ does not have this property since in a suspension ΣX all cup products of positive-dimensional classes are zero, according to an exercise for §3.2.

The fact that Steenrod squares are stable operations extending the cup product square yields the following theorem, which implies that the stable homotopy groups of spheres π_1^s, π_3^s, and π_7^s are nontrivial:

Theorem 4L.2. *If $f : S^{2n-1} \to S^n$ has Hopf invariant 1, then $[f] \in \pi_{n-1}^s$ is nonzero, so the iterated suspensions $\Sigma^k f : S^{2n+k-1} \to S^{n+k}$ are all homotopically nontrivial.*

Proof: Associated to a map $f : S^\ell \to S^m$ is the mapping cone $C_f = S^m \cup_f e^{\ell+1}$ with the cell $e^{\ell+1}$ attached via f. Assuming f is basepoint-preserving, we have the relation $C_{\Sigma f} = \Sigma C_f$ where Σ denotes reduced suspension.

If $f : S^{2n-1} \to S^n$ has Hopf invariant 1, then by (5), $Sq^n : H^n(C_f;\mathbb{Z}_2) \to H^{2n}(C_f;\mathbb{Z}_2)$ is nontrivial. By (4) the same is true for $Sq^n : H^{n+k}(\Sigma^k C_f;\mathbb{Z}_2) \to H^{2n+k}(\Sigma^k C_f;\mathbb{Z}_2)$ for all k. If $\Sigma^k f$ were homotopically trivial we would have a retraction $r : \Sigma^k C_f \to S^{n+k}$. The diagram at the right would then commute by naturality of Sq^n, but since the group in the lower left corner of the diagram is zero, this gives a contradiction. □

$$\begin{tikzcd} H^{n+k}(S^{n+k};\mathbb{Z}_2) \arrow[r, "r^*", "\approx"'] \arrow[d, "Sq^n"] & H^{n+k}(\Sigma^k C_f;\mathbb{Z}_2) \arrow[d, "Sq^n"] \\ H^{2n+k}(S^{n+k};\mathbb{Z}_2) \arrow[r, "r^*"] & H^{2n+k}(\Sigma^k C_f;\mathbb{Z}_2) \end{tikzcd}$$

The Steenrod power operations $P^i : H^n(X;\mathbb{Z}_p) \to H^{n+2i(p-1)}(X;\mathbb{Z}_p)$ for p an odd prime will satisfy analogous properties:

(1) $P^i(f^*(\alpha)) = f^*(P^i(\alpha))$ for $f : X \to Y$.

(2) $P^i(\alpha+\beta) = P^i(\alpha) + P^i(\beta)$.

(3) $P^i(\alpha \smile \beta) = \sum_j P^j(\alpha) \smile P^{i-j}(\beta)$ (the Cartan formula).

(4) $P^i(\sigma(\alpha)) = \sigma(P^i(\alpha))$ where $\sigma : H^n(X;\mathbb{Z}_p) \to H^{n+1}(\Sigma X;\mathbb{Z}_p)$ is the suspension isomorphism given by reduced cross product with a generator of $H^1(S^1;\mathbb{Z}_p)$.

(5) $P^i(\alpha) = \alpha^p$ if $2i = |\alpha|$, and $P^i(\alpha) = 0$ if $2i > |\alpha|$.

(6) $P^0 = \mathbb{1}$, the identity.

The germinal property $P^i(\alpha) = \alpha^p$ in (5) can only be expected to hold for even-dimensional classes α since for odd-dimensional α the commutativity property of cup product implies that $\alpha^2 = 0$ with $\mathbb{Z}_p$ coefficients if p is odd, and then $\alpha^p = 0$ since $\alpha^2 = 0$. Note that the formula $P^i(\alpha) = \alpha^p$ for $|\alpha| = 2i$ implies that P^i raises dimension by $2i(p-1)$, explaining the appearance of this number.

The Bockstein homomorphism $\beta : H^n(X;\mathbb{Z}_p) \to H^{n+1}(X;\mathbb{Z}_p)$ is not included as one of the P^i's, but this is mainly a matter of notational convenience. As we shall see later when we discuss Adem relations, the operation Sq^{2i+1} is the same as the composition $Sq^1 Sq^{2i} = \beta Sq^{2i}$, so the Sq^{2i}'s can be regarded as the P^i's for $p = 2$.

One might ask if there are elements of π_*^s detectable by Steenrod powers in the same way that the Hopf maps are detected by Steenrod squares. The answer is yes for the operation P^1, as we show in Example 4L.6. It is a perhaps disappointing fact that no other squares or powers besides Sq^1, Sq^2, Sq^4, Sq^8, and P^1 detect elements of homotopy groups of spheres. (Sq^1 detects a map $S^n \to S^n$ of degree 2.) We will prove this for certain Sq^i's and P^i's later in this section. The general case for $p = 2$ is Adams' theorem on the Hopf invariant discussed in §4.B, while the case of odd p is proved in [Adams & Atiyah 1966]; see also [VBKT].

The Cartan formulas can be expressed in a more concise form by defining *total* Steenrod square and power operations by $Sq = Sq^0 + Sq^1 + \cdots$ and $P = P^0 + P^1 + \cdots$. These act on $H^*(X;\mathbb{Z}_p)$ since by property (5), only a finite number of Sq^i's or P^i's can be nonzero on a given cohomology class. The Cartan formulas then say that $Sq(\alpha \smile \beta) = Sq(\alpha) \smile Sq(\beta)$ and $P(\alpha \smile \beta) = P(\alpha) \smile P(\beta)$, so Sq and P are ring homomorphisms.

We can use Sq and P to compute the operations Sq^i and P^i for projective spaces and lens spaces via the following general formulas:

$$(*) \qquad \begin{aligned} Sq^i(\alpha^n) &= \binom{n}{i}\alpha^{n+i} \ \text{ for } \alpha \in H^1(X;\mathbb{Z}_2) \\ P^i(\alpha^n) &= \binom{n}{i}\alpha^{n+i(p-1)} \ \text{ for } \alpha \in H^2(X;\mathbb{Z}_p) \end{aligned}$$

To derive the first formula, properties (5) and (6) give $Sq(\alpha) = \alpha + \alpha^2 = \alpha(1+\alpha)$, so $Sq(\alpha^n) = Sq(\alpha)^n = \alpha^n(1+\alpha)^n = \sum_i \binom{n}{i}\alpha^{n+i}$ and hence $Sq^i(\alpha^n) = \binom{n}{i}\alpha^{n+i}$. The second formula is obtained in similar fashion: $P(\alpha) = \alpha + \alpha^p = \alpha(1+\alpha^{p-1})$ so $P(\alpha^n) = \alpha^n(1+\alpha^{p-1})^n = \sum_i \binom{n}{i}\alpha^{n+i(p-1)}$.

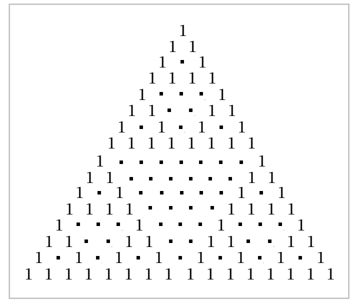
In Lemma 3C.6 we described how binomial coefficients can be computed modulo a prime p:

$$\binom{m}{n} \equiv \prod_i \binom{m_i}{n_i} \bmod p, \qquad \text{where } m = \sum_i m_i p^i \text{ and } n = \sum_i n_i p^i \text{ are the } p\text{-adic expansions of } m \text{ and } n.$$

When $p = 2$ for example, the extreme cases of a dyadic expansion consisting of a single 1 or all 1's give

$$Sq(\alpha^{2^k}) = \alpha^{2^k} + \alpha^{2^{k+1}}$$
$$Sq(\alpha^{2^k-1}) = \alpha^{2^k-1} + \alpha^{2^k} + \alpha^{2^k+1} + \cdots + \alpha^{2^{k+1}-2}$$

for $\alpha \in H^1(X;\mathbb{Z}_2)$. More generally, the coefficients of $Sq(\alpha^n)$ can be read off from the $(n+1)$ st row of the mod 2 Pascal triangle, a portion of which is shown in the figure at the right, where dots denote zeros.



Example 4L.3: Stable Splittings. The formula $(*)$ tells us how to compute Steenrod squares for $\mathbb{R}\mathrm{P}^\infty$, hence also for any suspension of $\mathbb{R}\mathrm{P}^\infty$. The explicit formulas for $Sq(\alpha^{2^k})$ and $Sq(\alpha^{2^k-1})$ above show that all the powers of the generator $\alpha \in H^1(\mathbb{R}\mathrm{P}^\infty;\mathbb{Z}_2)$ are tied together by Steenrod squares since the first formula connects α inductively to all the powers α^{2^k} and the second formula connects these powers to all the other powers. This shows that no suspension $\Sigma^k\mathbb{R}\mathrm{P}^\infty$ has the homotopy type of a wedge sum $X \vee Y$ with both X and Y having nontrivial cohomology. In the case of $\mathbb{R}\mathrm{P}^\infty$ itself we could have deduced this from the ring structure of $H^*(\mathbb{R}\mathrm{P}^\infty;\mathbb{Z}_2) \approx \mathbb{Z}_2[\alpha]$, but cup products become trivial in a suspension.

The same reasoning shows that $\mathbb{C}\mathrm{P}^\infty$ and $\mathbb{H}\mathrm{P}^\infty$ have no nontrivial stable splittings. The $\mathbb{Z}_2$ cohomology in these cases is again $\mathbb{Z}_2[\alpha]$, though with α no longer 1-dimensional. However, we still have $Sq(\alpha) = \alpha + \alpha^2$ since these spaces have no nontrivial cohomology in the dimensions between α and α^2, so we have $Sq^{2i}(\alpha^n) = \binom{n}{i}\alpha^{n+i}$ for $\mathbb{C}\mathrm{P}^\infty$ and $Sq^{4i}(\alpha^n) = \binom{n}{i}\alpha^{n+i}$ for $\mathbb{H}\mathrm{P}^\infty$. Then the arguments from the real case carry over using the operations Sq^{2i} and Sq^{4i} in place of Sq^i.

Suppose we consider the same question for $K(\mathbb{Z}_3,1)$ instead of $\mathbb{R}\mathrm{P}^\infty$. Taking cohomology with $\mathbb{Z}_3$ coefficients, the Bockstein β is nonzero on odd-dimensional classes in $H^*(K(\mathbb{Z}_3,1);\mathbb{Z}_3)$, thus tying them to the even-dimensional classes, so we only need to see which even-dimensional classes are connected by P^i's. The even-dimensional part of $H^*(K(\mathbb{Z}_3,1);\mathbb{Z}_3)$ is a polynomial algebra $\mathbb{Z}_3[\alpha]$ with $|\alpha| = 2$, so we have $P^i(\alpha^n) = \binom{n}{i}\alpha^{n+i(p-1)} = \binom{n}{i}\alpha^{n+2i}$ by our earlier formula. Since P^i raises dimension by $4i$ when $p = 3$, there is no chance that all the even-dimensional cohomology will be connected by the P^i's. In fact, we showed in Proposition 4I.3 that $\Sigma K(\mathbb{Z}_3,1) \simeq X_1 \vee X_2$ where X_1 has the cohomology of $\Sigma K(\mathbb{Z}_3,1)$ in dimensions congruent to 2 and 3 mod 4, while X_2 has the remaining cohomology. Thus the best one could hope would be that all the odd powers of α are connected by P^i's and likewise all the even powers are connected, since this would imply that neither X_1 nor X_2 splits nontrivially. This is indeed the case, as one sees by an examination of the coefficients in the formula $P^i(\alpha^n) = \binom{n}{i}\alpha^{n+2i}$. In the Pascal triangle mod 3, shown at the right, $P(\alpha^n)$ is determined by the $(n+1)$ st row. For example the sixth row says that $P(\alpha^5) = \alpha^5 + 2\alpha^7 + \alpha^9 + \alpha^{11} + 2\alpha^{13} + \alpha^{15}$. A few moments' thought shows that the rows that compute $P(\alpha^n)$ for $n = 3^k m - 1$ have all nonzero entries, and these rows together with the rows right after them suffice to connect the powers of α in the desired way, so X_1 and X_2 have no stable splittings. One can also see that $\Sigma^2 X_1$ and X_2 are not homotopy equivalent, even stably, since the operations P^i act differently in the two spaces. For example P^2 is trivial on suspensions of α but not on suspensions of α^2.

1
1 1
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The situation for $K(\mathbb{Z}_p, 1)$ for larger primes p is entirely similar, with $\Sigma K(\mathbb{Z}_p, 1)$ splitting as a wedge sum of $p-1$ spaces. The same arguments work more generally for $K(\mathbb{Z}_{p^i}, 1)$, though for $i > 1$ the usual Bockstein β is identically zero so one has to use instead a Bockstein involving $\mathbb{Z}_{p^i}$ coefficients. We leave the details of these arguments as exercises.

Example 4L.4: Maps of $\mathbb{H}\mathrm{P}^\infty$. We can use the operations P^i together with a bit of number theory to demonstrate an interesting distinction between $\mathbb{H}\mathrm{P}^\infty$ and $\mathbb{C}\mathrm{P}^\infty$, namely, we will show that if a map $f : \mathbb{H}\mathrm{P}^\infty \to \mathbb{H}\mathrm{P}^\infty$ has $f^*(\gamma) = d\gamma$ for γ a generator of $H^4(\mathbb{H}\mathrm{P}^\infty; \mathbb{Z})$, then the integer d, which we call the *degree* of f, must be a square. By contrast, since $\mathbb{C}\mathrm{P}^\infty$ is a $K(\mathbb{Z}, 2)$, there are maps $\mathbb{C}\mathrm{P}^\infty \to \mathbb{C}\mathrm{P}^\infty$ carrying a generator $\alpha \in H^2(\mathbb{C}\mathrm{P}^\infty; \mathbb{Z})$ onto any given multiple of itself. Explicitly, the map $z \mapsto z^d$, $z \in \mathbb{C}$, induces a map f of $\mathbb{C}\mathrm{P}^\infty$ with $f^*(\alpha) = d\alpha$, but commutativity of $\mathbb{C}$ is needed for this construction so it does not extend to the quaternionic case.

We shall deduce the action of Steenrod powers on $H^*(\mathbb{H}\mathrm{P}^\infty; \mathbb{Z}_p)$ from their action on $H^*(\mathbb{C}\mathrm{P}^\infty; \mathbb{Z}_p)$, given by the earlier formula $(*)$ which says that $P^i(\alpha^n) = \binom{n}{i}\alpha^{n+i(p-1)}$ for α a generator of $H^2(\mathbb{C}\mathrm{P}^\infty; \mathbb{Z}_p)$. There is a natural quotient map $\mathbb{C}\mathrm{P}^\infty \to \mathbb{H}\mathrm{P}^\infty$ arising from the definition of both spaces as quotients of S^∞. This map takes the 4-cell of $\mathbb{C}\mathrm{P}^\infty$ homeomorphically onto the 4-cell of $\mathbb{H}\mathrm{P}^\infty$, so the induced map on cohomology sends a generator $\gamma \in H^4(\mathbb{H}\mathrm{P}^\infty; \mathbb{Z}_p)$ to α^2, hence γ^n to α^{2n}. Thus the formula $P^i(\alpha^{2n}) = \binom{2n}{i}\alpha^{2n+i(p-1)}$ implies that $P^i(\gamma^n) = \binom{2n}{i}\gamma^{n+i(p-1)/2}$. For example, $P^1(\gamma) = 2\gamma^{(p+1)/2}$.

Now let $f : \mathbb{H}\mathrm{P}^\infty \to \mathbb{H}\mathrm{P}^\infty$ be any map. Applying the formula $P^1(\gamma) = 2\gamma^{(p+1)/2}$ in two ways, we get

$$\begin{aligned} &P^1 f^*(\gamma) = f^* P^1(\gamma) = f^*(2\gamma^{(p+1)/2}) = 2d^{(p+1)/2}\gamma^{(p+1)/2} \\ \text{and}\quad &P^1 f^*(\gamma) = P^1(d\gamma) = 2d\gamma^{(p+1)/2} \end{aligned}$$

Hence the degree d satisfies $d^{(p+1)/2} \equiv d \bmod p$ for all odd primes p. Thus either $d \equiv 0 \bmod p$ or $d^{(p-1)/2} \equiv 1 \bmod p$. In both cases d is a square mod p since the congruence $d^{(p-1)/2} \equiv 1 \bmod p$ is equivalent to d being a nonzero square mod p, the multiplicative group of nonzero elements of the field $\mathbb{Z}_p$ being cyclic of order $p-1$.

The argument is completed by appealing to the number theory fact that an integer which is a square mod p for all sufficiently large primes p must be a square. This can be deduced from quadratic reciprocity and Dirichlet's theorem on primes in arithmetic progressions as follows. Suppose on the contrary that the result is false for the integer d. Consider primes p not dividing d. Since the product of two squares in $\mathbb{Z}_p$ is again a square, we may assume that d is a product of distinct primes $q_1, \cdots, q_n$, where one of these primes is allowed to be -1 if d is negative. In terms of the Legendre symbol $(\frac{d}{p})$ which is defined to be $+1$ if d is a square mod p and -1 otherwise, we have

$$\left(\frac{d}{p}\right) = \left(\frac{q_1}{p}\right) \cdots \left(\frac{q_n}{p}\right)$$

The left side is $+1$ for all large p by hypothesis, so it will suffice to see that p can be chosen to give each term on the right an arbitrary preassigned value. The values of $(\frac{-1}{p})$ and $(\frac{2}{p})$ depend only on p mod 8, and the four combinations of values are realized by the four residues $1,3,5,7$ mod 8. Having specified the value of p mod 8, the quadratic reciprocity law then says that for odd primes q, specifying $(\frac{q}{p})$ is equivalent to specifying $(\frac{p}{q})$. Thus we need only choose p in the appropriate residue classes mod 8 and mod q_i for each odd q_i. By the Chinese remainder theorem, this means specifying p modulo 8 times a product of odd primes. Dirichlet's theorem guarantees that in fact infinitely many primes p exist satisfying this congruence condition.

It is known that the integers realizable as degrees of maps $\mathbb{H}P^\infty \to \mathbb{H}P^\infty$ are exactly the odd squares and zero. The construction of maps of odd square degree will be given in [SSAT] using localization techniques, following [Sullivan 1974]. Ruling out nonzero even squares can be done using K–theory; see [Feder & Gitler 1978], which also treats maps $\mathbb{H}P^n \to \mathbb{H}P^n$.

The preceding calculations can also be used to show that every map $\mathbb{H}P^n \to \mathbb{H}P^n$ must have a fixed point if $n > 1$. For, taking $p = 3$, the element $P^1(\gamma)$ lies in $H^8(\mathbb{H}P^n;\mathbb{Z}_3)$ which is nonzero if $n > 1$, so, when the earlier argument is specialized to the case $p = 3$, the congruence $d^{(p+1)/2} \equiv d \bmod p$ becomes $d^2 = d$ in $\mathbb{Z}_3$, which is satisfied only by 0 and 1 in $\mathbb{Z}_3$. Hence $f^*:\widetilde{H}^*(\mathbb{H}P^n;\mathbb{Z}_3) \to \widetilde{H}^*(\mathbb{H}P^n;\mathbb{Z}_3)$ is either zero or the identity. In both cases the Lefschetz number $\lambda(f)$, which is the sum of the traces of the maps $f^*:H^{4i}(\mathbb{H}P^n;\mathbb{Z}_3) \to H^{4i}(\mathbb{H}P^n;\mathbb{Z}_3)$, is nonzero, so the Lefschetz fixed point theorem gives the result.

Example 4L.5: Vector Fields on Spheres. Let us now apply Steenrod squares to determine the maximum number of orthonormal tangent vector fields on a sphere in all cases except when the dimension of the sphere is congruent to -1 mod 16. The first step is to rephrase the question in terms of Stiefel manifolds. Recall from the end of §3.D and Example 4.53 the space $V_{n,k}$ of orthonormal k-frames in $\mathbb{R}^n$. Projection of a k-frame onto its first vector gives a map $p:V_{n,k} \to S^{n-1}$, and a section for this projection, that is, a map $f: S^{n-1} \to V_{n,k}$ such that $pf = 1\!\!1$, is exactly a set of $k-1$ orthonormal tangent vector fields $v_1, \cdots, v_{k-1}$ on S^{n-1} since f assigns to each $x \in S^{n-1}$ an orthonormal k-frame $(x, v_1(x), \cdots, v_{k-1}(x))$.

We described a cell structure on $V_{n,k}$ at the end of §3.D, and we claim that the $(n-1)$-skeleton of this cell structure is $\mathbb{R}P^{n-1}/\mathbb{R}P^{n-k-1}$ if $2k-1 \le n$. The cells of $V_{n,k}$ were products $e^{i_1} \times \cdots \times e^{i_m}$ with $n > i_1 > \cdots > i_m \ge n-k$, so the products with a single factor account for all of the $(2n-2k)$-skeleton, hence they account for all of the $(n-1)$-skeleton if $n-1 \le 2n-2k$, that is, if $2k-1 \le n$. The cells that are products with a single factor are the homeomorphic images of cells of $\mathbb{R}P^{n-1}$ under a map $\mathbb{R}P^{n-1} \to SO(n) \to SO(n)/SO(n-k) = V_{n,k}$. This map collapses $\mathbb{R}P^{n-k-1}$ to a point, so we get the desired conclusion that $\mathbb{R}P^{n-1}/\mathbb{R}P^{n-k-1}$ is the $(n-1)$-skeleton of $V_{n,k}$ if $2k-1 \le n$.

Now suppose we have $f: S^{n-1} \to V_{n,k}$ with $pf = 1\!\!1$. In particular, f^* is surjective on $H^{n-1}(-;\mathbb{Z}_2)$. If we deform f to a cellular map, with image in the $(n-1)$-skeleton, then by the preceding paragraph this will give a map $g: S^{n-1} \to \mathbb{R}P^{n-1}/\mathbb{R}P^{n-k-1}$ if $2k-1 \le n$, and this map will still induce a surjection on $H^{n-1}(-;\mathbb{Z}_2)$, hence an isomorphism. If the number k happens to be such that $\binom{n-k}{k-1} \equiv 1 \bmod 2$, then by the earlier formula $(*)$ the operation

$$Sq^{k-1}: H^{n-k}(\mathbb{R}P^{n-1}/\mathbb{R}P^{n-k-1};\mathbb{Z}_2) \to H^{n-1}(\mathbb{R}P^{n-1}/\mathbb{R}P^{n-k-1};\mathbb{Z}_2)$$

will be nonzero, contradicting the existence of the map g since obviously the operation $Sq^{k-1}: H^{n-k}(S^{n-1};\mathbb{Z}_2) \to H^{n-1}(S^{n-1};\mathbb{Z}_2)$ is zero.

In order to guarantee that $\binom{n-k}{k-1} \equiv 1 \bmod 2$, write $n = 2^r(2s+1)$ and choose $k = 2^r + 1$. Assume for the moment that $s \ge 1$. Then $\binom{n-k}{k-1} = \binom{2^{r+1}s-1}{2^r}$, and in view of the rule for computing binomial coefficients in $\mathbb{Z}_2$, this is nonzero since the dyadic expansion of $2^{r+1}s-1$ ends with a string of 1's including a 1 in the single digit where the expansion of 2^r is nonzero. Note that the earlier condition $2k-1 \le n$ is satisfied since it becomes $2^{r+1}+1 \le 2^{r+1}s + 2^r$ and we assume $s \ge 1$.

Summarizing, we have shown that for $n = 2^r(2s+1)$, the sphere S^{n-1} cannot have 2^r orthonormal tangent vector fields if $s \ge 1$. This is also trivially true for $s = 0$ since S^{n-1} cannot have n orthonormal tangent vector fields.

It is easy to see that this result is best possible when $r \le 3$ by explicitly constructing $2^r - 1$ orthonormal tangent vector fields on S^{n-1} when $n = 2^r m$. When $r = 1$, view S^{n-1} as the unit sphere in $\mathbb{C}^m$, and then $x \mapsto ix$ defines a tangent vector field since the unit complex numbers 1 and i are orthogonal and multiplication by a unit complex number is an isometry of $\mathbb{C}$, so x and ix are orthogonal in each coordinate of $\mathbb{C}^m$, hence are orthogonal. When $r = 2$ the same construction works with $\mathbb{H}$ in place of $\mathbb{C}$, using the maps $x \mapsto ix$, $x \mapsto jx$, and $x \mapsto kx$ to define three orthonormal tangent vector fields on the unit sphere in $\mathbb{H}^m$. When $r = 3$ we can follow the same